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ICON Central Office - Madhapur - Hyderabad

**SEC: Sr.S60 (N&S), Elite, IAS & LIIT** **JEE-MAIN-Shift-2** **Date: 24-03-2026**

**Time: 03.00Pm to 06.00Pm** **GTM-27** **Max. Marks: 300**

**IMPORTANT INSTRUCTION:**

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   **(I) Section –I** contains **20 Multiple Choice Questions** with only one correct option.
   **Marking scheme: +4** for correct answer, **0** if not attempt and **-1** in all other cases.
   **(II) Section-II** contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



**Question Answered for Marking** **Question Cancelled for Marking**

**Marking scheme: +4** for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

**Name of the Candidate (in Capital): _____**

**Admission Number:** | | | | | | | | | |

**Candidate's Signature:_____** **Invigilator's Signature: _____**



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24-03-2026_ Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main-Shift-2_GTM-27_Test Syllabus

**MATHEMATICS : TOTAL SYLLABUS**

**PHYSICS : TOTAL SYLLABUS**

**CHEMISTRY : TOTAL SYLLABUS**

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**MATHEMATICS** **Max Marks: 100**

## SECTION-I
## (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.**

**1.** Let $m \in \mathbb{N}$, then $S = \sum_{k=1}^{m} k\left(\frac{1}{k} + \frac{1}{k+1} + \frac{1}{k+2} + ..... + \frac{1}{m}\right)$ is equal to

**1)** $\frac{1}{6}m(m+2)$ **2)** $\frac{1}{4}m(m+3)$ **3)** $\frac{1}{3}m(m+4)$ **4)** $\frac{1}{2}m(m+6)$

**2.** Let A= {1,2,3,4,5,6} and B = {1,2,3,4}, then number of functions

| | List – I | | List – II |
|---|---|---|---|
| P) | Defined from A to B and are either non – increasing or non – decreasing and into | **1)** | Is a prime number |
| Q) | Defined from A to B and are either non – increasing or non – decreasing and onto | **2)** | Is a perfect square |
| R) | Defined from B to B such that $f(i) \neq i\ \forall i \in B$ and into | **3)** | Is a multiple of 4 |
| S) | Defined from B to A such that $f(i) \neq i\ \forall i \in B$ and one –one | **4)** | Is having exactly 12 factors |
| | | 5) | Is a perfect cube |

The correct option is

**1)** P – (2,4), Q – (3), R– (1,5), S – (3)

**2)** P – (2, 3), Q – (3), R – (3, 4), S – (1)

**3)** P – (2, 4), Q – (5), R – (3,4), S – (3)

**4)** P – (2,3) , Q – (3) , R – (4, 5), S – (1)

**3.** The coordinates of the mid-points of the sides of $\Delta ABC$ are (3a, 0, 0), (0, 3b, 0) and (0, 0, 3c) respectively, then the area of $\Delta ABC$ is

**1)** $18\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$ **2)** $9\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$

**3)** $\frac{9}{2}\sqrt{b^2c^2 + c^2a^2 + a^2b^2}$ **4)** $18(ab + bc + ca)$



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**4.** Let $x_1, x_2, ...., x_n$ be $n$ observations. Let $w_i = lx_i + k$ for $i = 1, 2, ...., n$ where $l$ and k are constants. If the mean of $x_i's$ is 48 and their standard deviation is 12, the mean of $w_i's$ is 55 and standard deviation of $w_i's$ is 15, then the values of $l$ and $k$ can be

**1)** $l = \frac{5}{4}, k = -5$ **2)** $l = \frac{-5}{4}, k = 5$ **3)** $l = \frac{5}{2}, k = -5$ **4)** $l = \frac{5}{2}, k = 5$

**5.** Match the family of curves in column I with their differential equations in column II

| **Column – I** | **Column – II** |
|---|---|
| a) $y = ae^{3x} + be^{5x}$ | p) $y_2 + 16y = 0$ |
| b) $xy = ae^{x} + be^{-x} + x^2$ | q) $y_2 - 8y_1 + 15y = 0$ |
| c) $ax^2 + by^2 = 1$ | r) $xy_2 + 2y_1 - xy + x^2 - 2 = 0$ |
| d) $y = a\sin(4x + b)$ | s) $x(yy_2 + y_1^2) = yy_1$ |

**1)** $a-q, b-r, c-p, d-s$ **2)** $a-q, b-s, c-r, d-p$

**3)** $a-r, b-s, c-q, d-p$ **4)** $a-q, b-r, c-s, d-p$

**6.** Let $f:[0,1] \to [0,1]$ be a continuous function. Then,

**1)** $f(x) = x$ for at least one $x \in [0,1]$

**2)** $f(x) = x$ for at least one $x \in (0,1)$

**3)** $f(x) + x = 0$ for at least one $x \in [0,1]$

**4)** $f(x) + x = 0$ for at least one $x \in (0,1)$

**7.** The real values of '$a$' for which the point $(-2a, a+1)$ lies in the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$, is

**1)** $\left(-1, \frac{3}{5}\right)$ **2)** $\left(-1, -5 + 2\sqrt{6}\right)$

**3)** $\left(-5 - 2\sqrt{6}, -5 + 2\sqrt{6}\right)$ **4)** $\left(-5 - 2\sqrt{6}, \frac{3}{5}\right)$



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**8.** The length of the perpendicular from the origin to the plane passing through the point with position vector $\bar{a}$ and containing the line $\bar{r}=\bar{b}+\lambda\bar{c}$ is

**1)** $\left|\dfrac{[\bar{a}\bar{b}\bar{c}]}{|\bar{a}\times\bar{b}+\bar{b}\times\bar{c}+\bar{c}\times\bar{a}|}\right|$ **2)** $\left|\dfrac{[\bar{a}\bar{b}\bar{c}]}{|\bar{a}\times\bar{b}+\bar{b}\times\bar{c}|}\right|$

**3)** $\left|\dfrac{[\bar{a}\bar{b}\bar{c}]}{|(\bar{b}\times\bar{c})+(\bar{c}\times\bar{a})|}\right|$ **4)** $\left|\dfrac{[\bar{a}\bar{b}\bar{c}]}{|\bar{a}\times\bar{b}+\bar{c}\times\bar{a}|}\right|$

**9.** The maximum area of the rectangle whose sides pass through the vertices of a given the rectangle with sides a and b is

**1)** $\frac{1}{2}ab$ **2)** $\frac{1}{2}(a^2+b^2)$ **3)** $\frac{1}{2}(a+b)^2$ **4)** $\frac{1}{2}|a^2-b^2|$

**10.** In a planar rectangular coordinates system XOY, the area enclosed by the graph of function $f(x)=a\sin ax+\cos ax\ (a>0)$ defined on an interval with the least positive period and by the graph of function $g(x)=\sqrt{a^2+1}$ is

**1)** $\frac{\pi}{a}\sqrt{a^2-1}$ **2)** $\frac{2\pi}{a}\sqrt{a^2+1}$ **3)** $\frac{\pi}{2a}\sqrt{a^2+1}$ **4)** $\frac{4\pi}{a}\sqrt{a^2+1}$

**11.** Let D be the determinant given by

$$D=\begin{vmatrix}1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha)\\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta)\\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1\end{vmatrix}$$

Where $\alpha,\beta$ and $\gamma$ are real numbers.

**Statement – 1 :** The value of D is zero

**Statement – 2 :** The determinant D is expressible as the product of two determinants each equal to zero.

**1)** Statement-1 is true, statement-2 is true.

**2)** Statement-1 is false, statement-2 is true,

**3)** Statement-1 is true, Statement-2 is false

**4)** Statement-1 is false, Statement-2 is false.



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**12.** If $1,\alpha_1,\alpha_2,\alpha_3,\alpha_4$ are the fifth roots of unity, then $\sum_{r=1}^{4}\frac{1}{2-\alpha_r}=$

**1)** $\frac{51}{31}$ **2)** $\frac{49}{31}$ **3)** $\frac{25}{32}$ **4)** $\frac{25}{16}$

**13.** If $\frac{\tan 8^\circ}{1-3\tan^2 8^\circ}+\frac{3\tan 24^\circ}{1-3\tan^2 24^\circ}+\frac{9\tan 72^\circ}{1-3\tan^2 72^\circ}+\frac{27\tan 216^\circ}{1-3\tan^2 216^\circ}=x\tan 108^\circ+y\tan 8^\circ$

then $x+y=$

**1)** 8 **2)** 10 **3)** 6 **4)** 12

**14.** The value $\lim_{x\to 0}\frac{(1+x)^{1/x}-e+\frac{1}{2}ex}{x^2}$ is

**1)** $\frac{11e}{24}$ **2)** $\frac{e}{2}$ **3)** $\frac{e}{6}$ **4)** $\frac{7e}{12}$

**15.** The sum of the coefficients of those terms in the expansion of $(x_1+x_2+x_3+x_4)^8$ which Contain $x_3$ but not $x_4$ is ____

**1)** $4^8$ **2)** $2^8$ **3)** $3^8$ **4)** $3^8-2^8$

**16.** Minimum value of $27^{\cos x}+81^{\sin x}$ is equal to

**1)** $\frac{2\sqrt{3}}{9}$ **2)** $\frac{2\sqrt{3}}{27}$ **3)** $\frac{4\sqrt{3}}{9}$ **4)** $\frac{4\sqrt{3}}{27}$

**17.** $\int\frac{x^4-x^2}{(x^3+x)\sqrt{x^6+x^2}}dx=$

**1)** $\sec^{-1}\left(\frac{x^2+1}{\sqrt{2x}}\right)+c$ **2)** $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2x}}\right)+c$

**3)** $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}}\right)+c$ **4)** $\sec^{-1}\left(\frac{x^2-1}{\sqrt{2x}}\right)+c$

**18.** The value of $\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}+\tan^{-1}\frac{1}{21}+.....\infty$, is

**1)** $\frac{\pi}{2}$ **2)** $\frac{\pi}{4}$ **3)** $\frac{2\pi}{3}$ **4)** 0



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**19.** A hyperbola having transverse axis of length $2\sin\theta$ is confocal with the ellipse $3x^2+4y^2=12$, then its equation is

**1)** $x^2\cos ec^2\theta - y^2\sec^2\theta = 1$ **2)** $x^2 sec^2\theta - y^2\,\text{cosec}^2\,\theta = 1$

**3)** $x^2\sin^2\theta - y^2\cos^2\theta = 1$ **4)** $x^2\cos^2\theta - y^2\sin^2\theta = 1$

**20.** If $I_1=\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx$ and $I_2=\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx$, then $\frac{I_1}{I_2}$ equals

**1)** 2 **2)** $\frac{1}{\sqrt{2}}$ **3)** $\sqrt{2}$ **4)** $\frac{1}{2}$

## SECTION-II
## (NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

**Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.**

**21.** If $f$ is a real valued function satisfying the relation $1+f(x+y)\le f(x)+f(y)\le \sin x+\sin y+2, \forall x,y$ then f is a periodic function with fundamental period $k\pi$ then k is equal to_____

**22.** Let 'm' be the real solution to the equation $29\left(\frac{17-7x}{x+2}+\frac{8x+55}{x+3}\right)=31\left(\frac{24-5x}{x+1}+\frac{5-6x}{x+4}\right)+370$. There are positive integers a & b such that $m=-\frac{a}{b}$ then the minimum value of $a^b$ is _____

**23.** A square is inscribed in the circle $x^2+y^2-10x-6y+30=0$. One side of this square is parallel to $y=x+3$. If $(x_i, y_i)$ where $i=1,2,3,4$ are the vertices of the square, then $\sum_{i=1}^{4}(x_i+y_i)$ is equal to

**24.** If $S=\{1,2,3.....100\}$. Three numbers a, b, c are selected at random (with replacement) form set S. If largest number which appears is at least 81, then the probability that smallest number which appears is at most 40 is $\frac{p}{q}$;( $p,q\in N$ *and* q is a prime number), then p + q is equal to_____

**25.** Let A and B be two non-singular matrices such that $A\neq I$, $B^3=I$ and $AB=BA^2$, where I is the identify matrix, the least value of $k\in\mathbb{N}$ such that $A^k=I$ is _____



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**PHYSICS** **Max Marks: 100**

## SECTION-I
## (SINGLE CORRECT ANSWER TYPE)

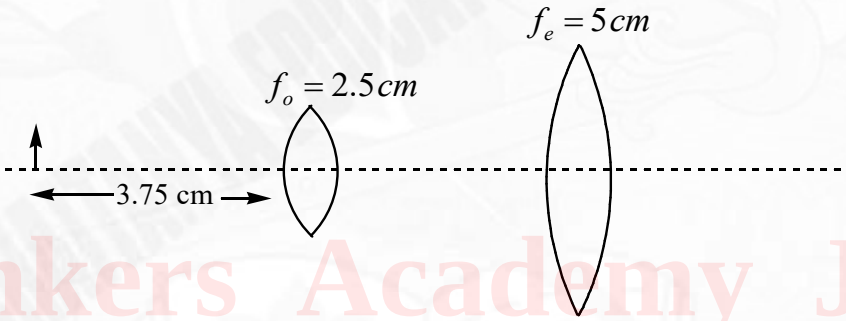
This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.**

**26.** Three capacitors each of capacitance 'C' and of breakdown voltage 'V' are joined in series. The capacitance and break down voltage of the series combination will be

**1)** $\frac{C}{3},\frac{V}{3}$ **2)** $3C,\frac{V}{3}$ **3)** $\frac{C}{3},3V$ **4)** 3C, 3V

**27.** In a compound microscope the focal length of the objective and the eye lens are 2.5 cm and 5 cm respectively. An object is placed at 3.75 cm before the objective lens and final image is formed at the least distance of distinct vision as shown in figure (Least distance of distinct vision is 25 cm) Which of the following statement is correct?




**1)** The magnitude of magnification of objective lens is 2

**2)** The magnitude of magnification of objective lens is 3

**3)** The magnitude of magnification of objective lens is 4

**4)** The magnitude of magnification of objective lens is 5

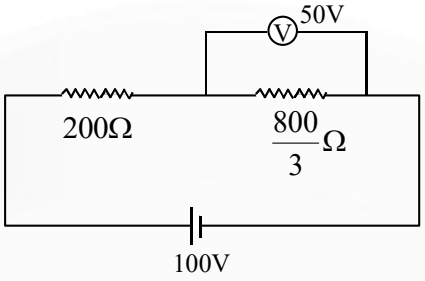
**28.** An electric bulb is designed to draw power (average power in one time period) at R.M.S voltage $V_0$. If the R.M.S voltage is 'V' it draws a power (Average power in one time period] P, then

**1)** $P=\left(\frac{V_0}{V}\right)^2 P_0$ **2)** $P=\left(\frac{V}{V_0}\right)^2 P_0$ **3)** $P=\left(\frac{V}{V_0}\right)P_0$ **4)** $P=\left(\frac{V_0}{V}\right)P_0$



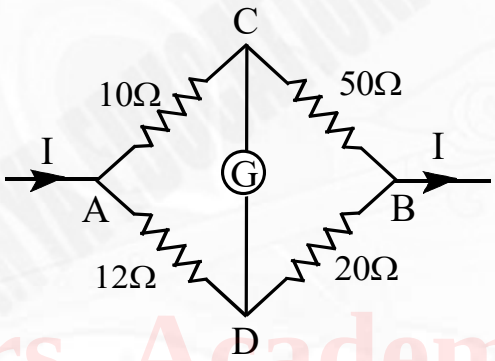
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**29.** What is the resistance of non-ideal voltmeter of reading is 50V shown in the circuit? Neglect the internal resistance of the Battery of e.m.f 100V



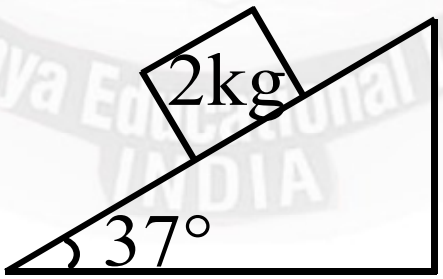

**1)** $\frac{800}{3}\Omega$ **2)** 800Ω **3)** 1000Ω **4)** 200Ω

**30.** In the situation shown, what is the value of resistance connected between A and D in parallel to 12Ω resistance, so that galvanometer shows no deflection?




**1)** 12Ω **2)** 6Ω **3)** 4Ω **4)** 5Ω

**31.** A block of mass 2kg is at rest on an fixed rough inclined plane of inclination angle 37° (tan37° = 3/4). Coefficient of friction between block, rough inclined plane (μ = 4/3). Calculate the force of friction acting on the block (take g = 10 $m/s^{2)}$ as shown in figure.

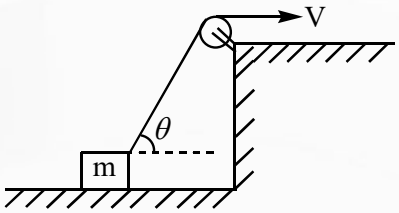



**1)** 20 N **2)** 12N **3)** 16 N **4)** Zero



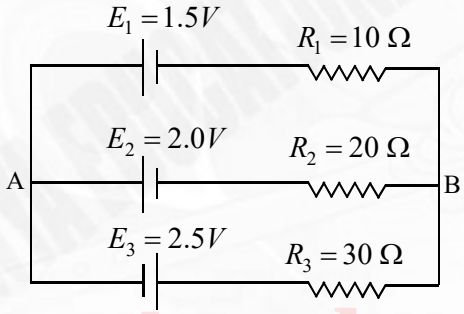
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**32.** A block of mass m is dragged on smooth horizontal plane with the help of rope which moves with velocity V. Neglect the friction between rope and fixed massless pulley. The horizontal velocity of the block is




**1)** V **2)** $\frac{V}{\sin\theta}$ **3)** $\frac{V}{\tan\theta}$ **4)** $\frac{V}{\cos\theta}$

**33.** In the circuit the sources have e.m.f's $E_1 = 1.5V, E_2 = 2.0V, E_3 = 2.5V$ and the resistances are equal to $R_1 = 10\Omega$, $R_2 = 20\Omega$, $R_3 = 30\Omega$. The internal resistances of the sources are negligible as shown in figure




Which of the following statement is correct

**1)** The modulus of current through $R_1 = 10\ \Omega$ resistance is $\frac{13}{220}A$

**2)** The modulus current through $R_1 = 10\ \Omega$ resistance is $\frac{13}{440}A$

**3)** The modulus of current through $R_1 = 10\ \Omega$ resistance is $\frac{13}{660}A$

**4)** The modulus current through $R_1 = 10\ \Omega$ resistance is $\frac{13}{110}A$

**34.** In an instrument there are 25 divisions on the vernier scale which have length of 24 divisions on the main scale. 1 cm on mains scale is divided in 20 equal parts. Find the least count.

**1)** 0.01 cm **2)** 0.005 cm **3)** 0.001 cm **4)** 0.002 cm



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**35.** An ammeter of resistance $3\Omega$ gives full scale deflection for 40 mA. What is the series resistance to be connected to convert the ammeter into a voltmeter of range 6V?

**1)** $147\Omega$ **2)** $153\Omega$ **3)** $63\Omega$ **4)** $36\Omega$

**36.** A point charge is projected at an initial velocity $\vec{u}$ in a uniform external magnetic field $\vec{B}$. $\theta$ is the angle between $\vec{u}$ and $\vec{B}$. Match the following columns (neglect gravitational effect)

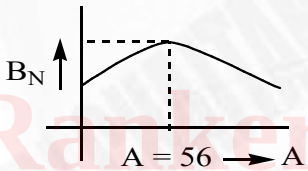
| **Angle '$\theta$'** | **Trajectory** |
|---|---|
| **1)** $\theta = 0$ | p) Parabola |
| **2)** $\theta = \frac{\pi}{2}$ | q) Ellipse |
| **3)** $\theta = \frac{\pi}{3}$ | r) Straight line |
| **4)** $\theta = \pi$ | s) Circle |
| | t) Helix |

**1)** 1 – s, 2 – t, 3 – p, 4 – s **2)** 1 – s, 2 – q, 3 – r, 4 – s

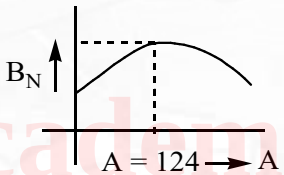
**3)** 1 – q, 2 – t, 3 – t, 4 – s **4)** 1 – r, 2 – s, 3 – t, 4 – r

**37.** In dependence of binding energy per nucleon, $B_N$ on the mass number, A is represented by

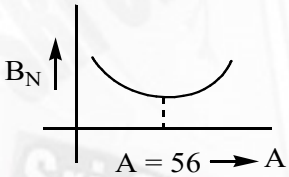
**1)**



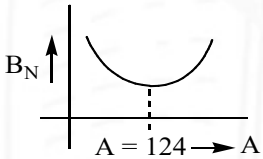

**2)**



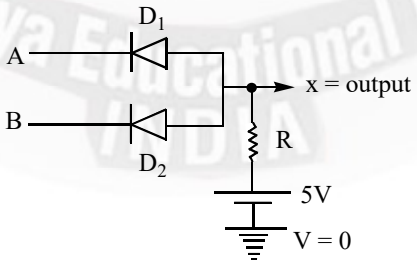

**3)**




**4)**




**38.** The circuit given below represents which of the logic operations (A and B are considered on if $V_A = V_B = 5V$) diodes are ideal

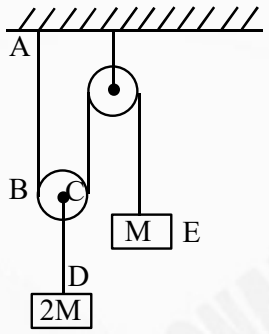



**1)** AND Gate **2)** NOT Gate **3)** OR Gate **4)** NOR Gate



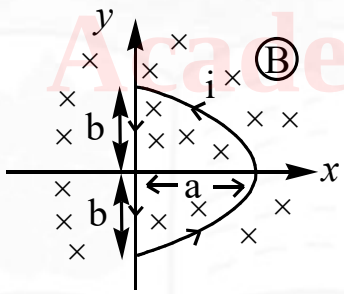
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**39.** Both the strings, shown in figure, are made of same material and have same cross-section. The pulleys and strings are light and the hanging blocks D, E are in equilibrium. The wave speed of a transverse wave in the string AB is $V_1$ and in CD string it is $V_2$. Then $\frac{V_1}{V_2}$ is




**1)** 1 **2)** 2 **3)** $\sqrt{2}$ **4)** $\frac{1}{\sqrt{2}}$

**40.** In the figure, there is a thin conducting wire having current I and which has a shape of closed half ellipse $\left[\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\right]$ is kept in a uniform magnetic field B as shown. The magnitude of magnetic dipole moment of loop and magnitude of torque acting on it are




**1)** $\pi abi, 0$ **2)** $\pi abi, \pi abiB$

**3)** $\frac{\pi ab}{2}i, \frac{\pi abiB}{2}$ **4)** $\frac{\pi abi}{2}, 0$

**41.** A series LCR circuit is at resonance for a capacitor of capacitance C, inductance L and resistance R. Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now

**1)** Zero **2)** Same **3)** Double **4)** Halved



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**42.** **Assertion :** The time period of revolution of a satellite (circular orbit) close to surface of earth is smaller than that revolving away from surface of earth.

**Reason :** The square of time period revolution of a satellite is directly proportional to cube of it's circular orbital radius

**1)** If both assertion and reason are true and reason is the correct explanation of assertion.

**2)** If both assertion and reason are true and reason is not the correct explanation of assertion

**3)** Assertion is true but reason is false

**4)** Both assertion and reason are false.

**43.** **Statement – I:** Kinetic energy of a particle is conserved if it is acted upon by a conservative force only

**Statement – II:** Work done by a conservative force in a closed path is zero

**1)** If both statements are true and the statement II is correct explanation of statement I

**2)** If both statements are true but statement II is not correct explanation of statement I

**3)** If statement I is true but the statements is false

**4)** If statement I is false but the statement II is true

**44.** Given below are two statements

**Statement – I:** Dimensional formula of plank's constant is $ML^2T^{-1}$.

**Statement – II:** Dimensional formula of bulk modulus is $ML^{-1}T^{-2}$.

**1)** Both the statements I and II are correct

**2)** Both the statements I and II are incorrect

**3)** Statement I is incorrect and statement II is correct

**4)** Statement I is correct and statement II is incorrect

**45.** Given below are two statements

**Statement – I:** Viscosity of gases is greater than that of liquids

**Statement – II:** Surface Tension of water decreases due to detergents.

**1)** Both the statements I and II are correct

**2)** Both the statements I and II are incorrect

**3)** Statement I is incorrect and statement II is correct

**4)** Statement I is correct and statement II is incorrect



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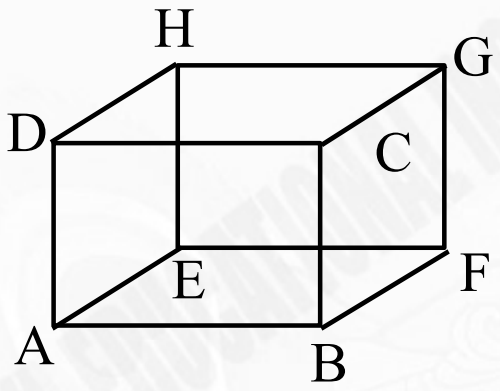

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## SECTION-II
## (NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

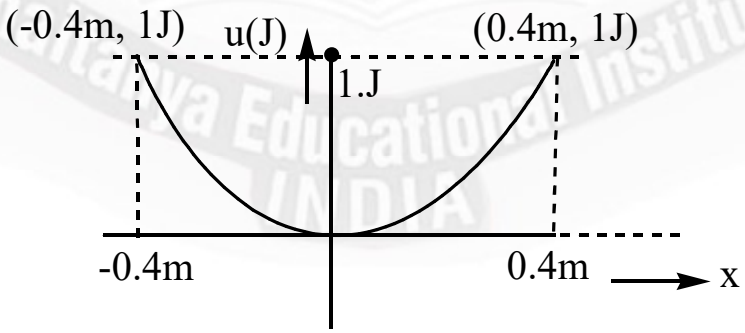
**Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases**

**46.** Consider an imaginary cube marked as shown case (I). A point positive charge is placed at vertex 'H' and flux through the face ABCD is $\phi$. Case II Now, the charge is shifted to center of the cube then flux through the face AEHD is K$\phi$. What is the value of K




**47.** An ideal monoatomic gas undergoes a process in which it's internal energy U and density $\rho$ vary as $U\rho =$ constant. The ratio of change in internal energy and the work done by the gas is (0.5x). Find the value of 'x'

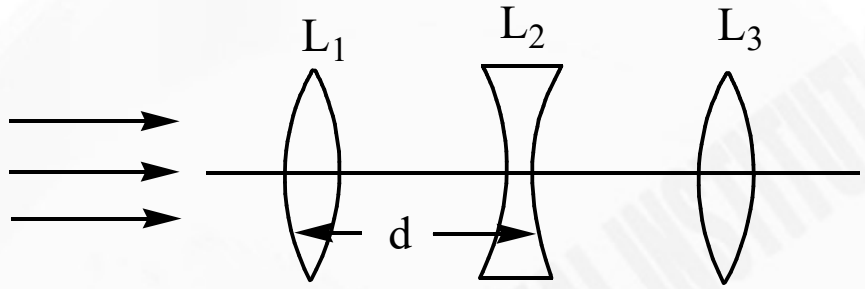
**48.** A particle of mass 2kg moves in simple harmonic motion and it's potential energy 'u' varies with position x as shown. The time period of oscillation of the particle is $\frac{n\pi}{5}$ seconds. Find the value of 'n'



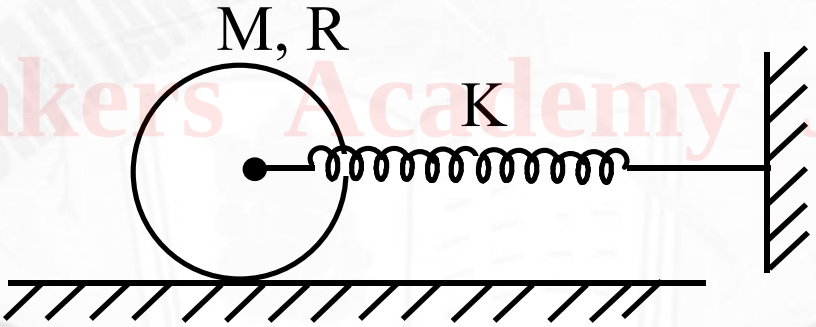



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**49.** Three thin lens $L_1$, $L_2$, $L_3$ are placed co-axially as shown in figure. Focal lengths of lenses are given magnitudes 30cm, 10cm, and 5cm respectively. If a parallel beam of (paraxial) light falling on lens $L_1$, emerging $L_3$ as a convergent beam such that it converges at the second focus of $L_3$.d distance between $L_1$ and $L_2$ will be (in centimeters)




**50.** A thin solid disc of mass M and radius R attached to horizontal massless spring of spring constant K can roll without slipping along the rough horizontal fixed surface as shown in figure. The time period of small oscillations of the center of mass of the disc is $\pi\sqrt{\frac{C.M}{K}}$ then value of C




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## CHEMISTRY

**Max Marks: 100**

### SECTION-I
### (SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.**

**51.** **Assertion (A):** Benzene is more stable than hypothetical cyclohexatriene

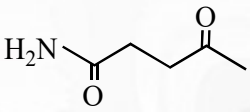
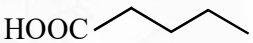
**Reason(R):** The delocalized π- electron cloud is attracted more strongly by nuclei of carbon atoms.

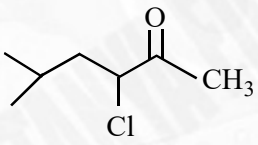
**1)** Both A and R correct and R is the correct explanation of A

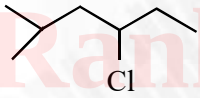
**2)** Both A and R correct But R is not the correct explanation of A

**3)** A is correct but R is incorrect

**4)** Both A and R incorrect

**52.** Statement – I  on Clemmensen reduction give 

Statement – II :  on Wolf Kishner reduction give



**1)** Statement I is true but statement II is false

**2)** Statement I is false but statement II is true

**3)** Statement I and statement II are false

**4)** Statement I and statement II are true

**53.** **Assertion(A):** Aniline is a better nucleophile than anilinium ion

**Reason(R):** Anilinium ion have positive charge

**1)** Both A and R correct and R is the correct explanation of A

**2)** Both A and R correct But R is not the correct explanation of A

**3)** A is correct but R is incorrect

**4)** Both A and R incorrect



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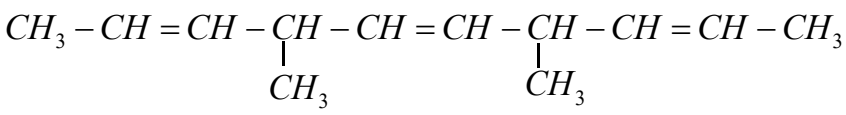

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**54.** Which of the following purification technique is used for separation of propanol and propanone?

**1)** Fractional distillation **2)** Simple distillation

**3)** Chromotography **4)** Differential extraction

**55.** IUPAC name of the following



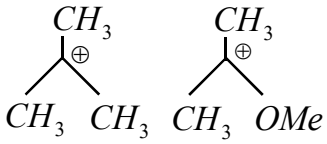
**1)** 4,7 dimethyl. Dec – 2, 5, 8 – triene

**2)** 5,8 diethyl. Dec – 2, 5, 8 – triene

**3)** 2,5 tetramethyl. Dec – 4, 7, 8 – triene

**4)** 4,7 trimethyl. Dec – 2, 5, 8 – triene

**56.**



A B

What is true for the following

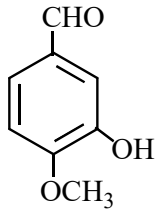
**1)** A is more stable than B

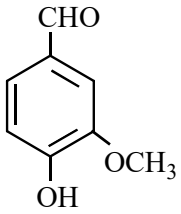
B) A has + R group – $CH_3$

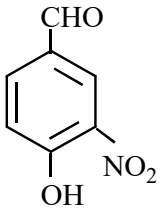
**3)** B has less stability due to –I effect of –$CH_3$

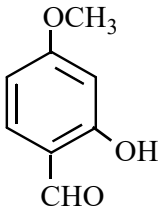
**4)** B has +M effect

**57.** Vanilin is

**1)** 

**2)** 

**3)** 

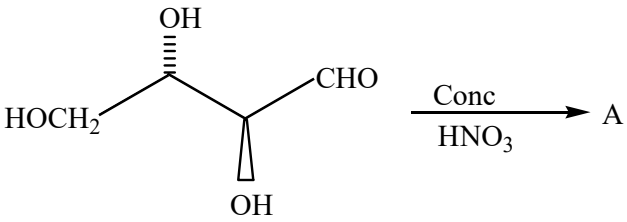
**4)** 

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**58.**




**1)** A is optically active racemic mixture

**2)** A is optically inactive meso compound

**3)** Only one compound optically active is formed

**4)** Both 1 and 2

**59.** **Statement I:** Soda bottles are sealed under high pressure

**Statement II :** High pressure decreases the solubility of gas in the liquid making it free from $CO_2$

**1)** Statement I is true and Statement II is false

**2)** Statement I is false and Statement II is true

**3)** Statement I and Statement II are false

**4)** Statement I and Statement II are true

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**60.** **Statement I :** During isobaric change $q_{rev} = q_{irrev}$

**Statement II :** During isobaric process, $\Delta S_{system} = -\Delta S_{surrounding}$

**1)** Statement I is true and Statement II is false

**2)** Statement I is false and Statement II is true

**3)** Statement I and Statement II are false

**4)** Statement I and Statement II are true

**61.** $Pt|H_2|H^+||H_2\ Pt$ is a cell diagram pH of anode and cathode are 7 and 9 respectively.

Check whether the cell will be spontaneous or not

**1)** Spontaneous **2)** Non Spontaneous

**3)** Equilibrium **4)** Can't Predict



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**62.** Calculate valence factor of the following reaction $FeS_2 + O_2 \rightarrow Fe_2O_3 + SO_2$

**1)** 12 **2)** 11 **3)** 8 **4)** 10

**63.** Hypo solution on treatment with free $I_2$ generates a product has how many linkages between same atom

**1)** 1 **2)** 3 **3)** 4 **4)** 10

**64.**

| $[A](M)$ | $[B](M)$ | $t_{1/2}(s)$ |
|---|---|---|
| 1 | 2 | 0.3 |
| 1 | 4 | 0.6 |
| 3 | 2 | 0.4 |
| 6 | 2 | 0.4 |

What is the order of reaction

**1)** 0 **2)** 1st **3)** 2nd **4)** Can't predict

**65.** $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$ if the pressure of the system is doubled what will be the effect flow of reaction?

**1)** Goes forward **2)** Goes backward

**3)** Same on both direction **4)** No effect

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**66.** How many moles of $OH^-$ will be required to neutralize 3 moles of $PCl_3$ on hydrolysis?

**1)** 6 **2)** 3 **3)** 9 **4)** 15

**67.** $NH_4Cl$ reacts with sodium nitrite What will be the number of anti-bonding electrons in the gaseous product formed after thermal decomposition.

**1)** 14 **2)** 10 **3)** 4 **4)** Zero

**68.** How many of the following compounds have an E – X – E bond angle more than 109°28' (E is the common element on both sides of X. X can be the same as E or different)

$H_2O, Ph-OH, Cr_2O_7^{2-}, I_3^{-1}, CH_3-O-CH_3$

**1)** 2 **2)** 3 **3)** 4 **4)** Zero

**69.** Which of the following element has lowest melting point in 3d series

**1)** Sc **2)** V **3)** Hg **4)** Zn



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**70.**

| **Comp** | **Shape** |
|---|---|
| a) $NO_2^-$ | p) Triangular |
| b) $CO_2$ | q) Linear |
| c) $SF_4$ | r) T Shape |
| d) $BrF_3$ | s) See – Saw |

**1)** a – p, b – q, c – s, d – r **2)** a – q, b – p, c – r, d – s

**3)** a – p, b – q, c – r, d – s **4)** a – s, b – q, c – r, d – p

## SECTION-II
## (NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

**Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases**

**71.** $[Cr(en)_3]^{3+}$ total isomers shown

**72.** During reaction of dil nitric acid with copper what is the oxidation state of nitrogen in a product which is in gaseous form

**73.** How many of the following have $Sp^3d$ hybridization

$PCl_5$, $BrF_3$, $SF_4$, $NH_3$, $H_2O$, ClF, $NO_3^{-1}$ , $SF_6$, $CH_4$, $CO_2$.

**74.** How many of the following contain oxy linkage between common elements $(E-O-E)$ linkage, Berylium acetate (alkaline), $H_2S_2O_7$, $H_2S_2O_8$, $H_4P_2O_6$, $N_2O_4$, $N_2O_3$, $N_2O_3$ (symmetrical), $Mn_2O_7$

**75.** Evaluate spin multiplicity of $Cr^{+3}$ ion.



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